Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 01



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Recap of Previous Lecture



- Topic — Doubly Linked list
- Topic — Circular Linked list
- Topic
- Topic
- Topic

Slide

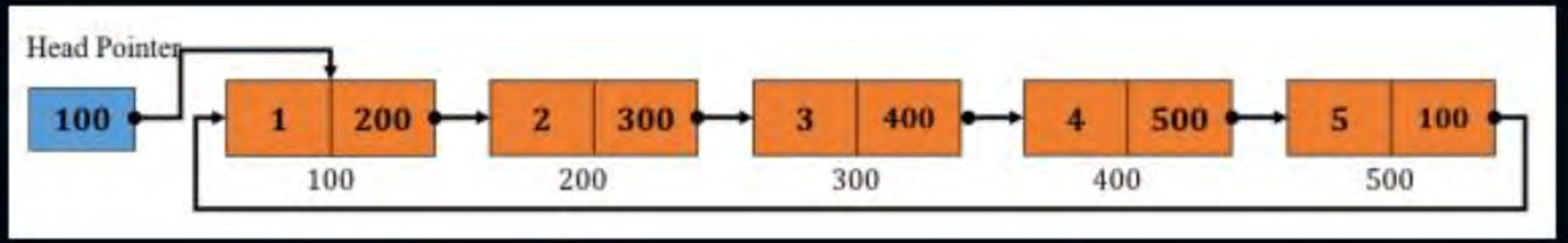
Topics to be Covered



- Topic — Circular Linked list
- Topic — Tree, general Tree
- Topic — forest
- Topic — k-ary Tree.
- Topic

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Topic : Circular Linked List



Insert x after y . Single linked

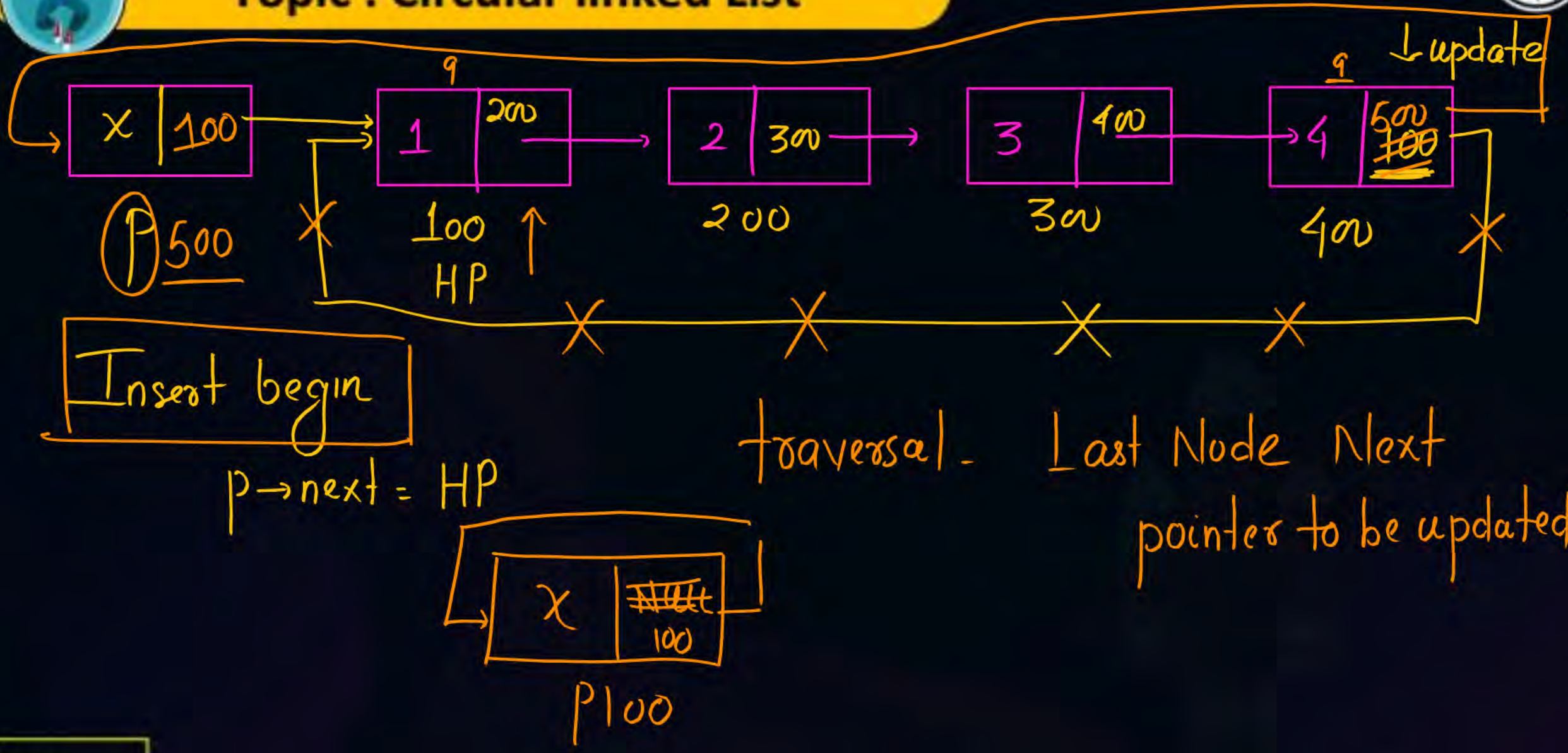
* Insert begin Insert end which one required traversal

Begin & end both required traversal

Slide

Topic : Circular linked List





Insert begin

P → next = HP

traversal - Last Node Next pointer to be updated

Slide



Topic : Circular linked List

HP global variable

```
void Addbegin(int x) {
    Node* p = getnode(x);
    Node* q = HP;
    if (HP == NULL) {
        HP = p
        p -> next = p
    }
```

Empty

Code

```
    else {
        while(q -> next != HP)
            q = q -> next;
        p -> next = HP;
        q -> next = p;
        HP = p; // Head pointer updated
    }
}
```

Slide



Topic : Circular linked List

HW : what modification needed in above code so that it will work Addend (Adding element at end of list)

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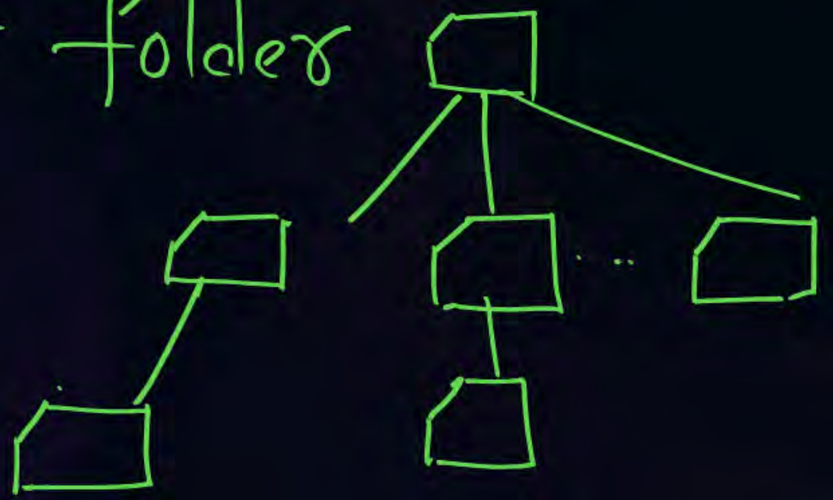
Topic : Tree

Non Linear Data structure (Tree)

* Recursion Linear (factorial)

  fibonacci series (Non Linear)

* folder



* organizaiton structure

Slide



Topic : Tree

Tree (Graph)

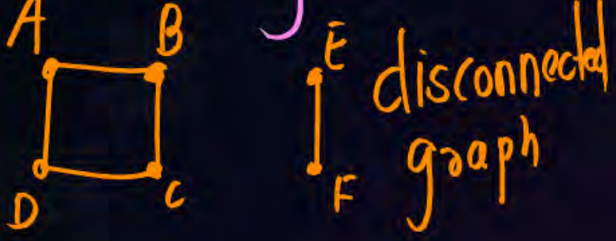
* Tree is set of Nodes & edges.

* Tree is acyclic graph (Tree does not contain cycle)



* Tree having n node always has n-1 edges

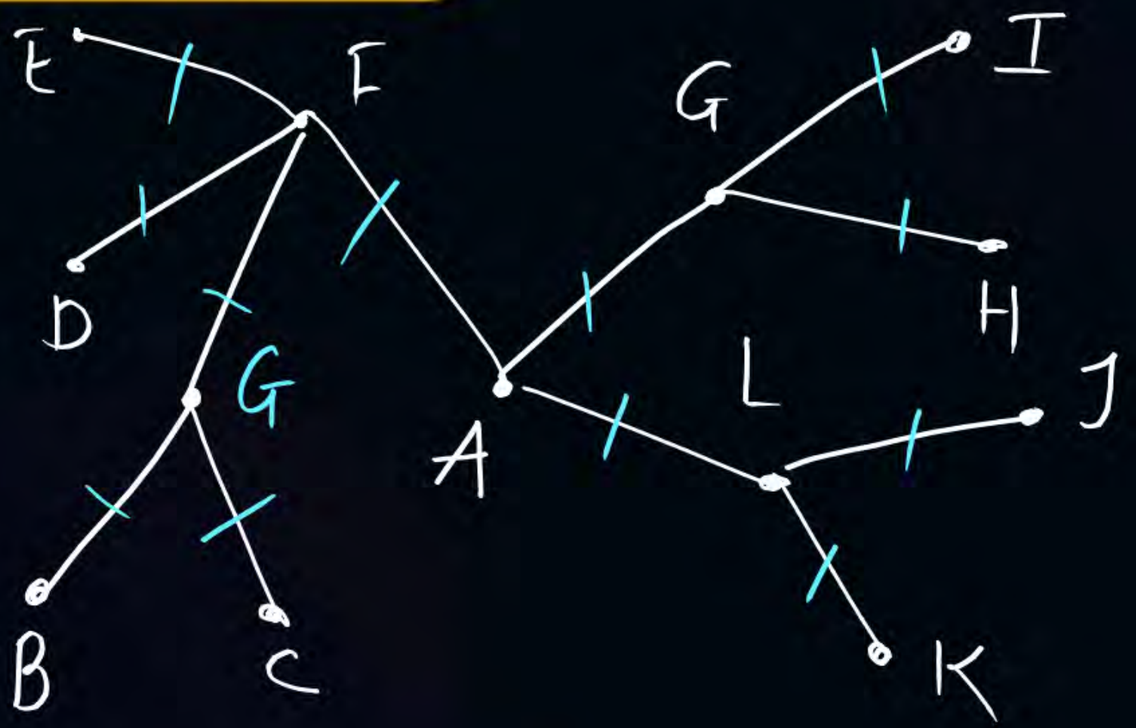
* Tree is a connected graph.



Slide



Topic : Tree



Acycle

Connected

n = 13

edges = 12

Spanning Tree

Slide



Topic : Tree

Rooted Tree : One Node is Selected as Root of tree other Nodes are arranged in parent children.

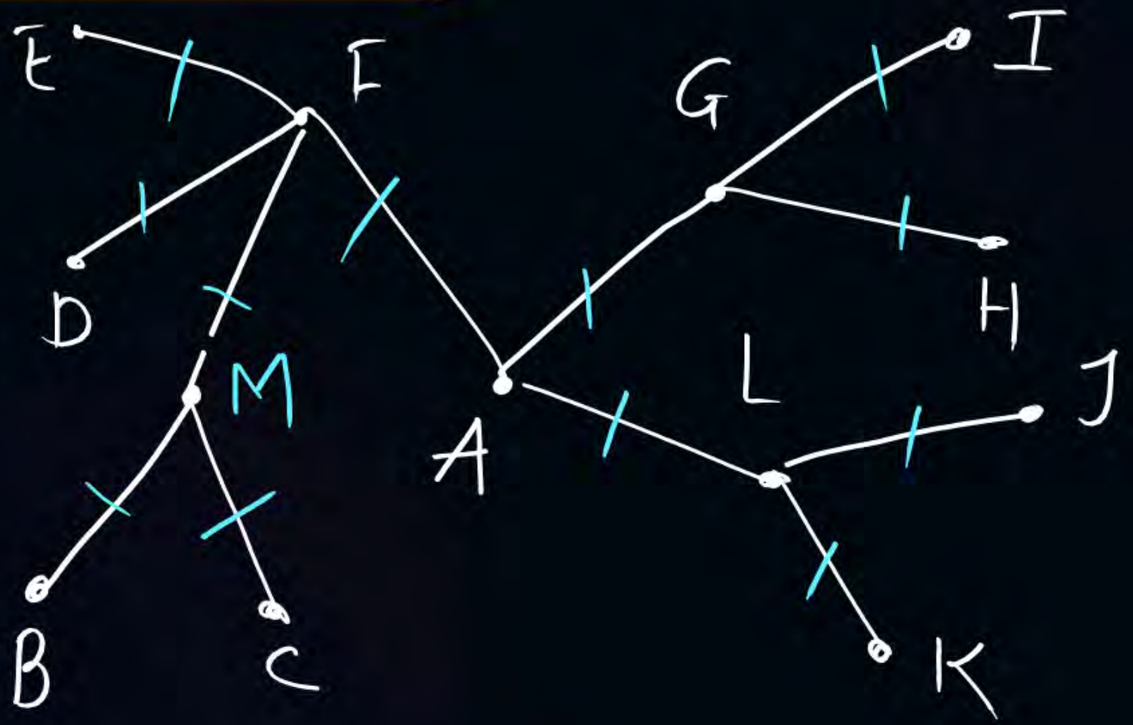
ONE DISTINGUISHED NODE Selected as Root.

∴ Root does Not have any parent.

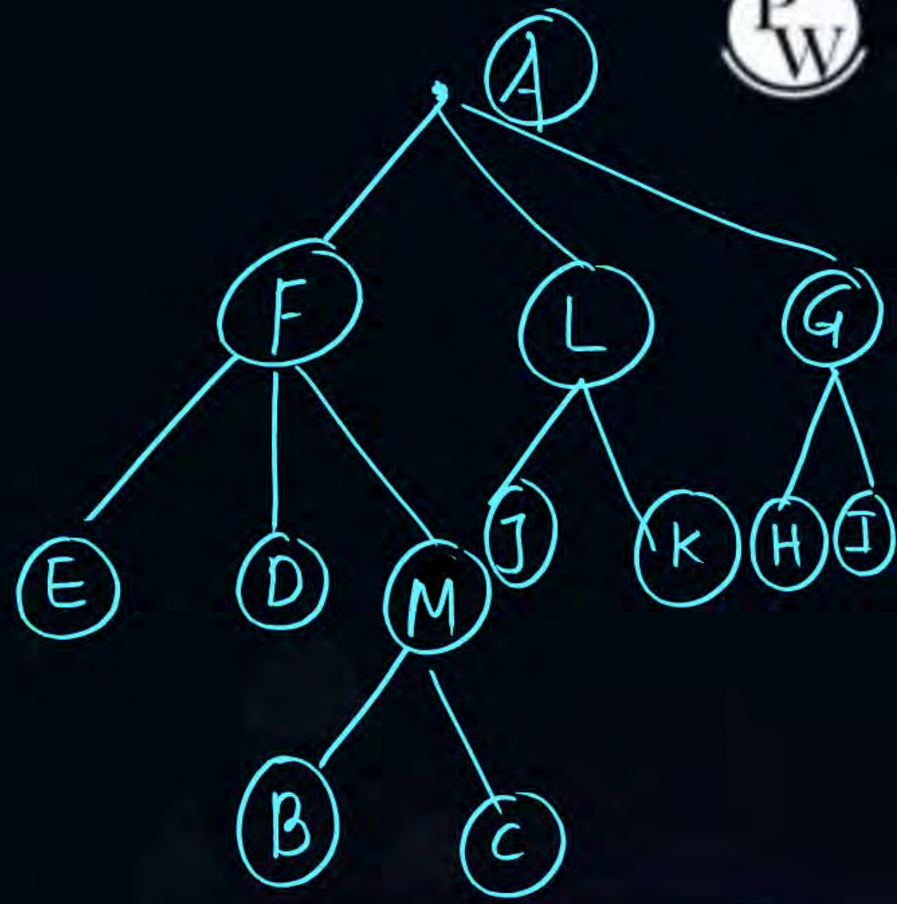
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Topic : Tree



Spanning Tree



Rooted Tree

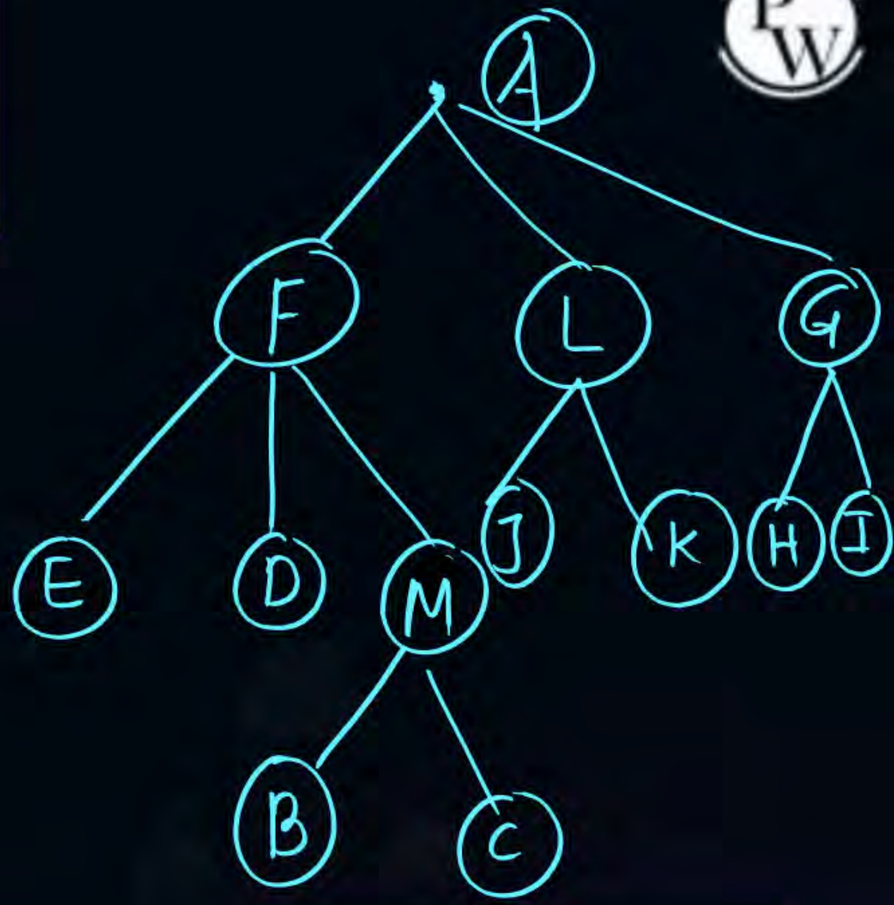
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Topic : Tree

Terminology

* A root of tree
* F is parent of E, D, M
* B, C are children of M
* E, D, B, C, J, K, H, I, leaf Node
* A, F, L, G, M Internal Node.
* Total No. of Nodes = # of leaf Node + # Internal Node



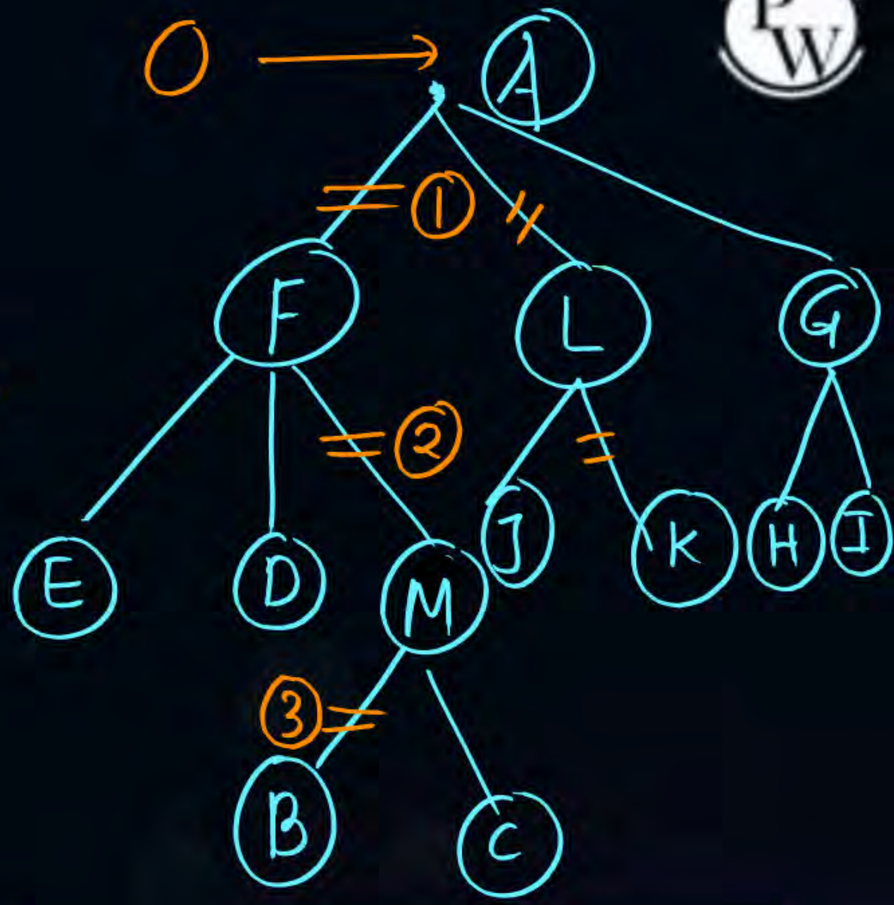
Rooted Tree

Slide



Topic : Tree

* Root Node is at height 0
* Height of tree is longest Root to Leaf path (# of edges)
* Height of tree is 3
* D is at Depth 2, G is at depth 1



Rooted Tree

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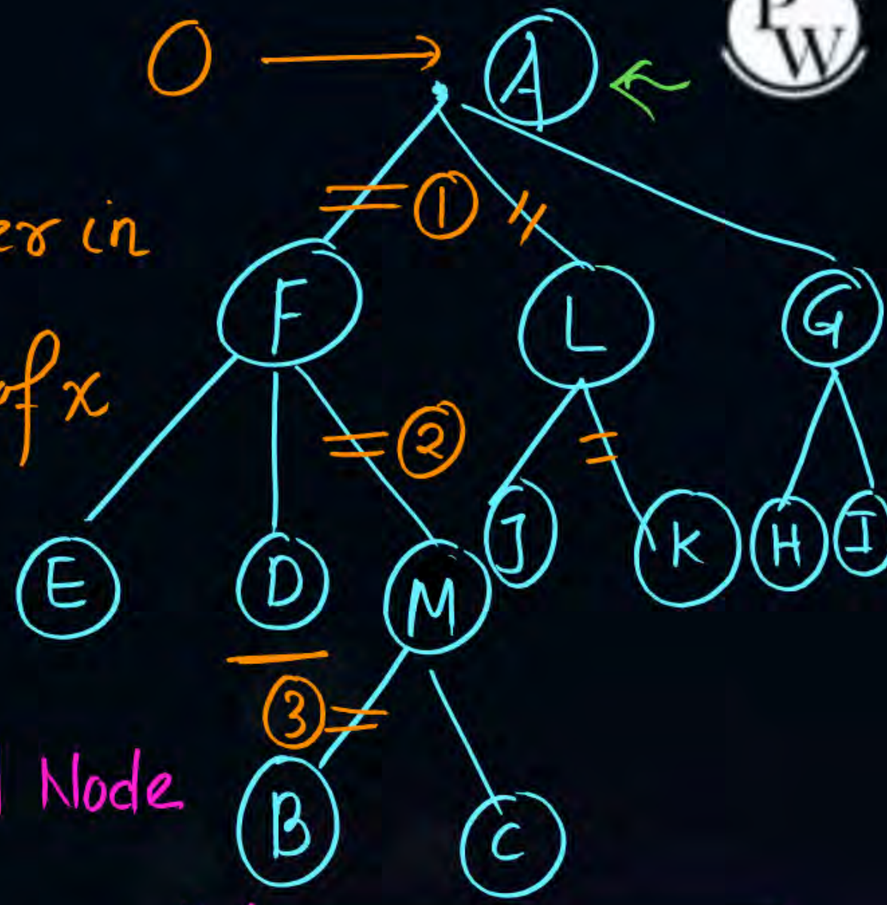


Topic : Tree

* **Ancestors** : All Node that encounter in Node x to root path called ancestors of x

F, A are ancestof of D

* **Descendant** : for given Node x all Node that encounter in Node x to leaf path are called Descendant of x. E, D, M, B, C are decendant of F

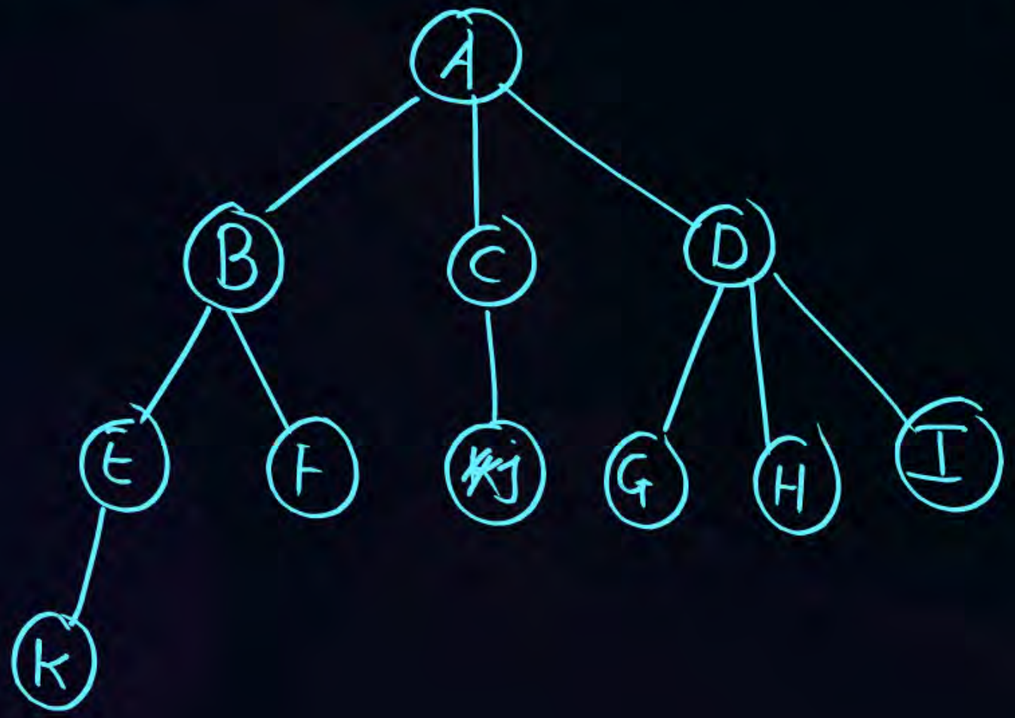


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Topic : Tree

General Tree : A tree is called general tree in which each Node is having 0 or more children.



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Topic : Tree

-Forest : forest is collection of one or more general Tree.

Each general tree is called component.

12 edges

16 - 4 = (12)



4 components in forest.

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Topic : Tree

* A tree having n node has (n-1) edges.

Theorem : No. of edges in forest having p components and n nodes total (Total No. of Node is n in forest)

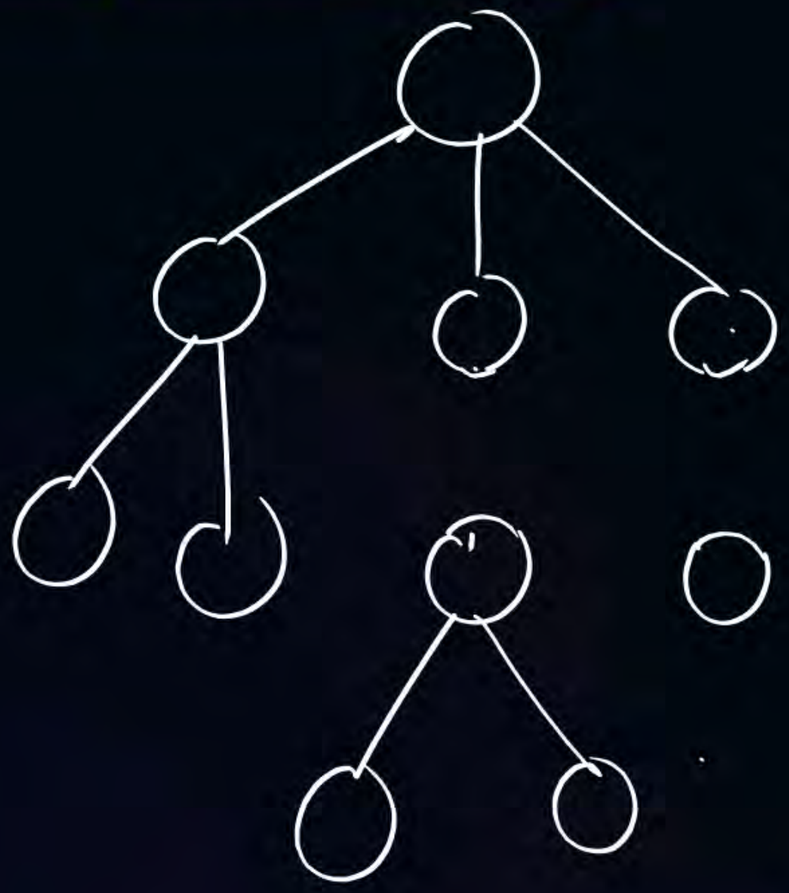
1992
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$n - p$ ✓

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Topic : Tree



if No. of Node is n, component 1 edges
n-1

after removing 1 edge. Nodes – n
Component : 2
edges - n-2

after removing 2 edges : Node - n
Component : 3
edges : n-3

p component :
(n-p)

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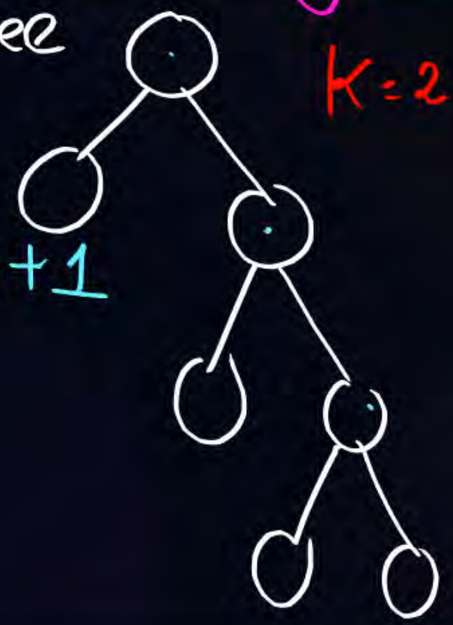
Topic : Tree

Special type

k-ary Tree : A tree is called k-ary tree if each Node is having 0 or k children.

2-ary tree

K=2

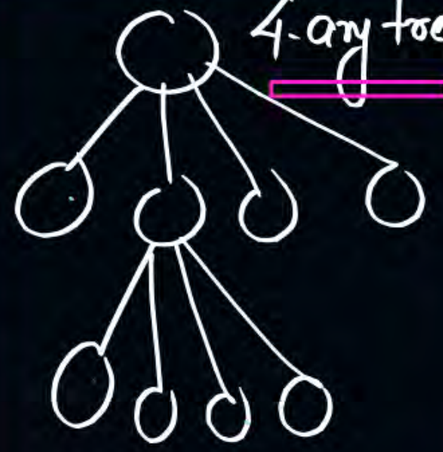


$L = 3 \times (2-1) + 1$

$= 3 + 1$

$= 4$

4-ary tree



$L = 2 \times (4-1) + 1$

$2 \times 3 + 1 = 7$



K=3

3-ary tree

$L = 3 \times (3-1) + 1$

$= 3 \times 2 + 1 = 7$

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Topic : Tree

Theorem : if L is No. of leaves & I is No. of Internal Node the

$$L = I(K-1) + 1$$

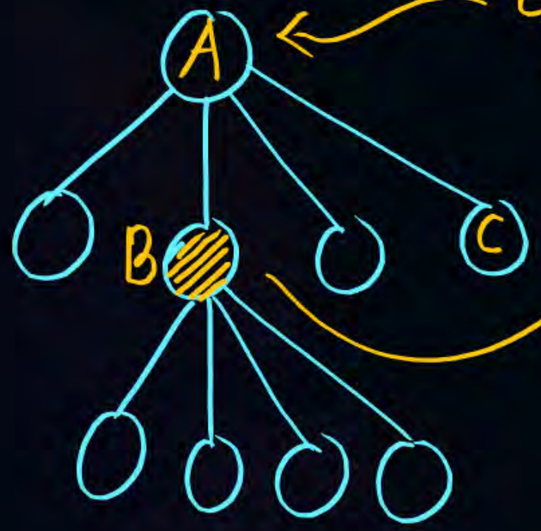
proof

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Topic : Tree

degree of Node : degree of a Node is No. of edges it connected to.



degree = 4

(B) degree of 5

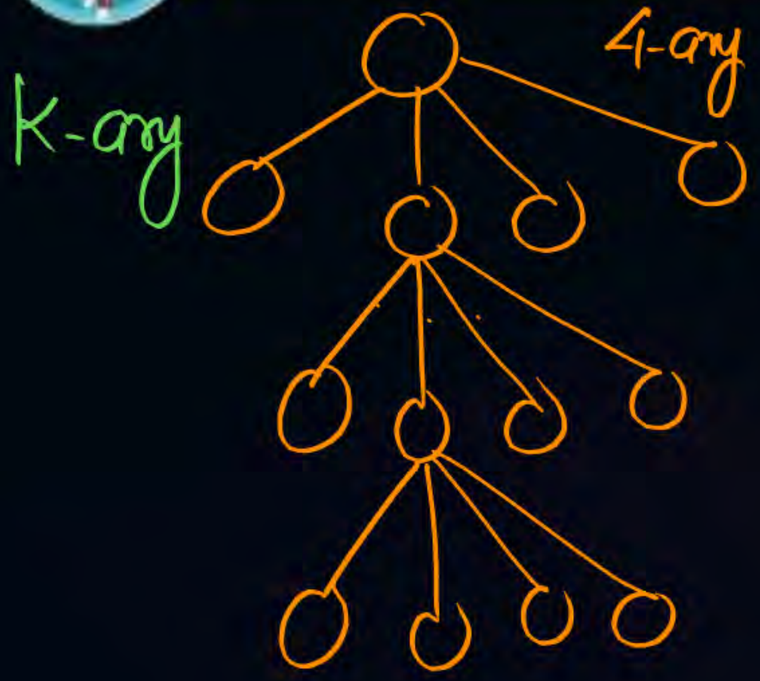
Leaf Nod degeeee = 1

Degree of (C) = 1

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Topic : Tree



L is leaf Node degree of (L) = 1

I is Internal Node, degree of Internal = k+1 (with 1 exception)

total No. of Nodes = L+I

total No. of edges = L+I-1

Handshaking Theorem

$$L \times 1 + I(k+1) - 1 = 2(L+I-1)$$

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Topic : Question

HW

A 2-3 tree is tree such that

(A) all internal nodes have either 2 or 3 children

(B) all paths from root to the leaves have the same length

The minimum and maximum number of internal nodes of a 2-3 tree having 9 leaves could be

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2 mins Summary

Topic — Circular Linked List

Topic — Tree Terminology

Topic — general Tree, forest

Topic — k-ary Tree

Topic

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THANK - YOU

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